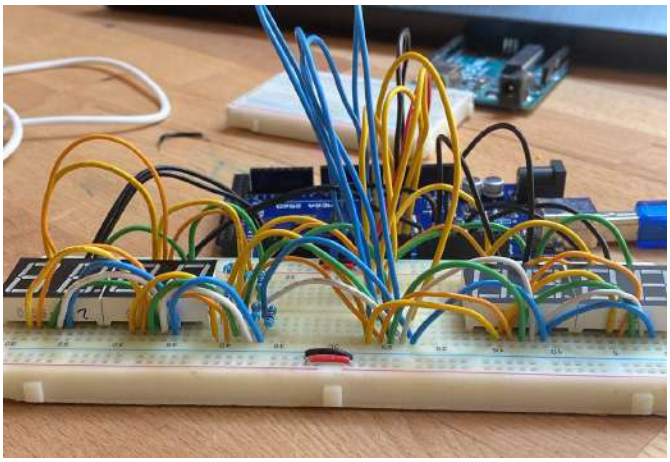
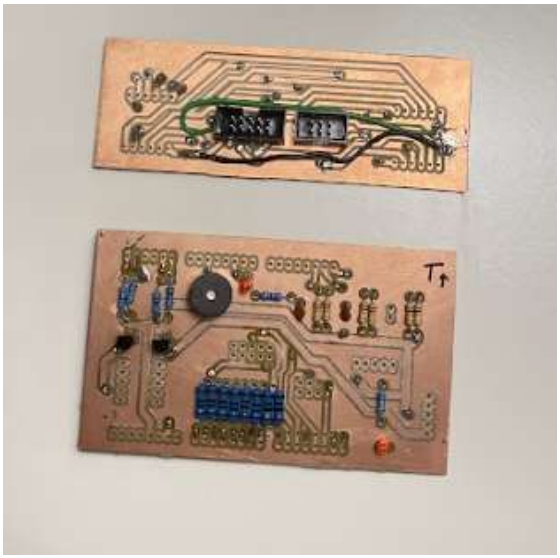
Project Portfolio - Bassam El Merhebi

Chess timer Project:

For this project, I wanted to make a chess game timer from scratch, instead of buying one using Arduino and components.



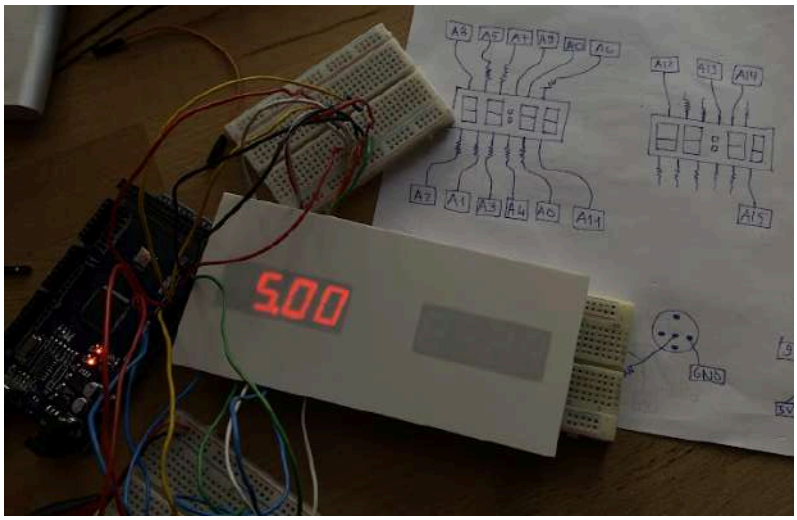
I started by wiring the 7-segment LED digits to an Arduino Mega.
I chose the mega, as the Arduino uno has only 20 pins I can use, and I did not want to use multiplexing to avoid complication.
8 pins for segments; 1 pin per digit; hence 8 digit pins; leaving me with only 4 pins for buzzer; leds; and buttons.
Breadboarding first to test everything is working



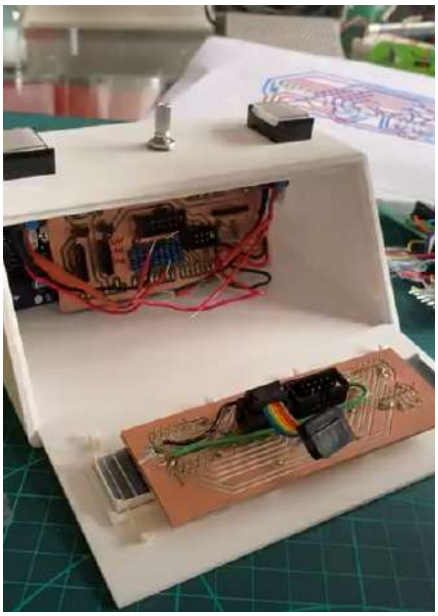
Designed the schematic in eagle and used my mini CNC machine to mill the double sided PCB to minimize the size of the timer, onto which I soldered the components.

Notice, the black and green jumper wires, could not route all the traces as I recall, and this was a solution.

Then came the aesthetics. I started with a rough prototype on how the chess timer was going to look and modeled it in Fusion 360 in order to 3D Print



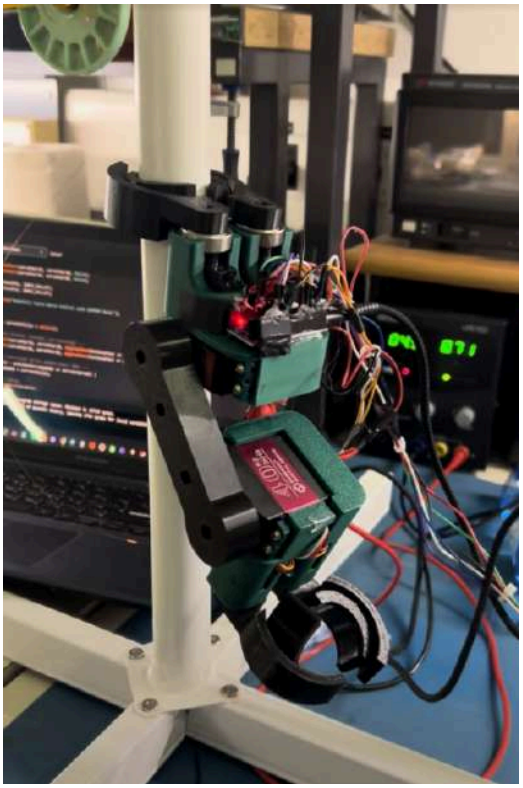
The first step was the front face. In the part of the face where the LED's sit thinner so the light shines through seamlessly, creating a smooth look.

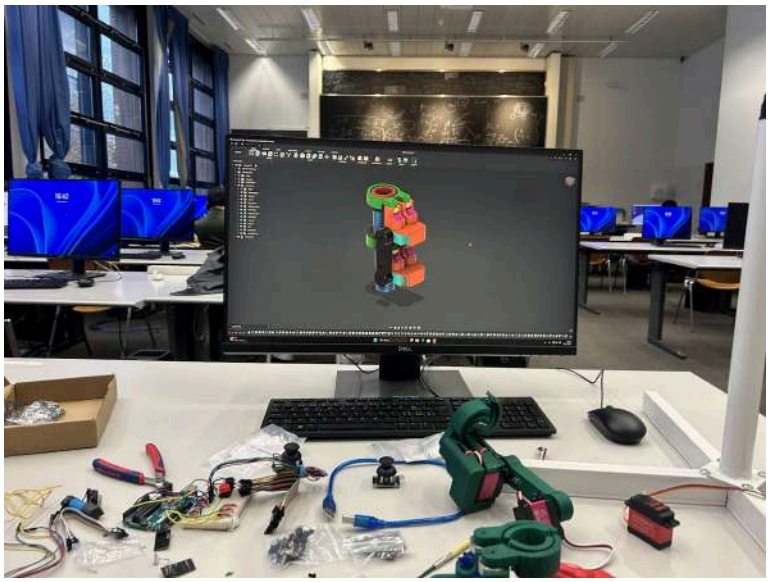


Then I printed the case, where I installed the PCB's, buttons, encoder, battery and the Arduino mega.

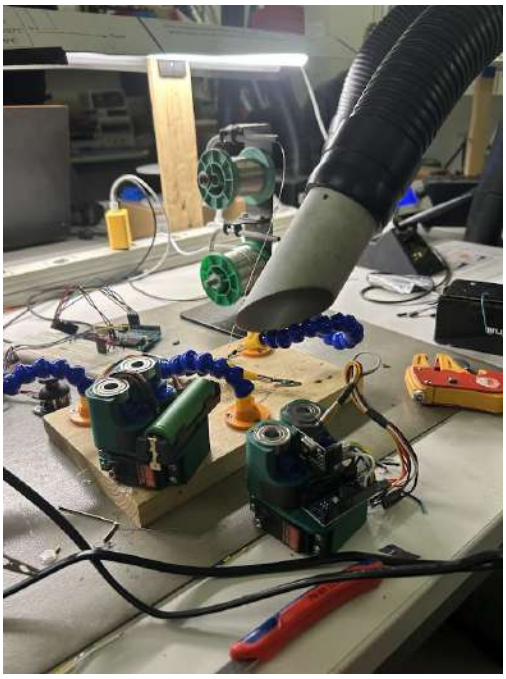
After closing the case and doing the battery work, here is the final product.

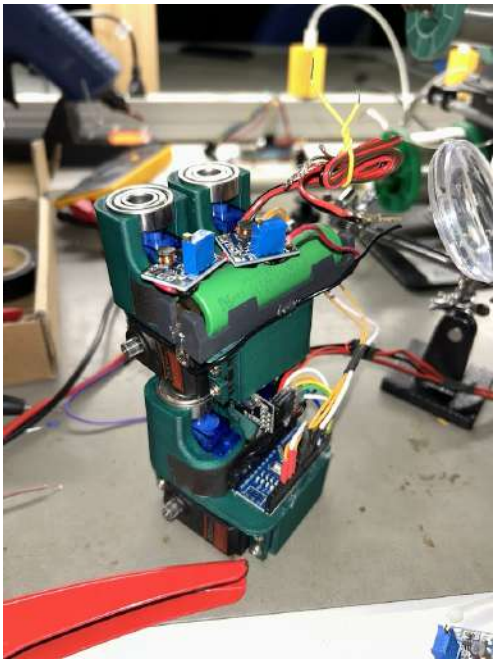
Tree Climber Robot

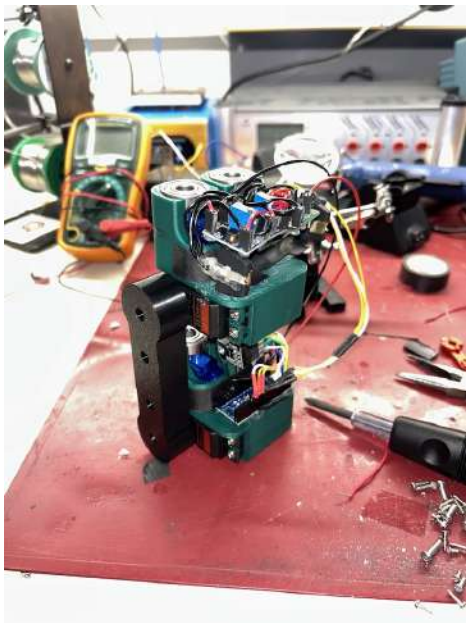




Project for the Swiss Robotics Day 2025: A tree climbing robot that perpetually hooks and flips around and climbs:
Arduino, servos, Fusion 360 CAD, power / batteries
A video is available on the GIT

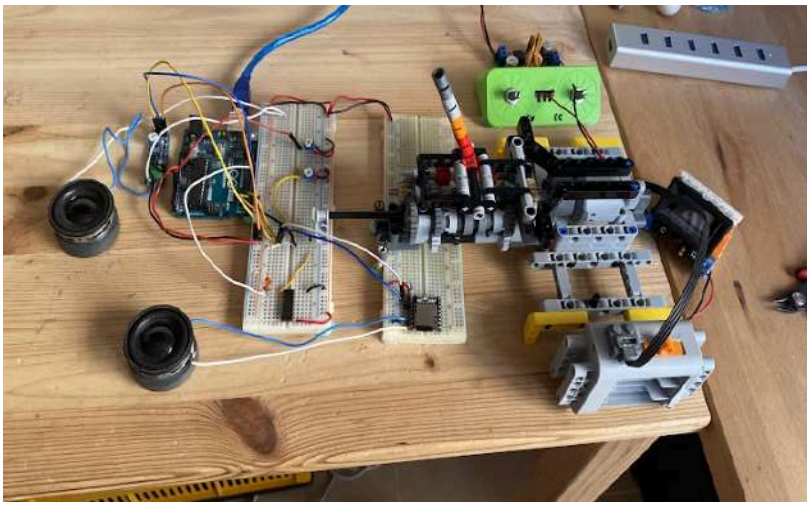






Car sound bike emulator Project:

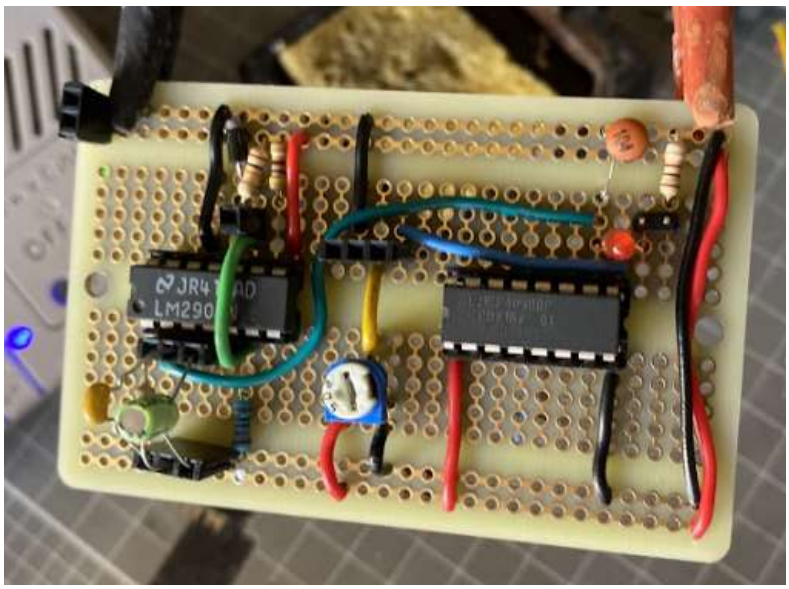
For context, I love car sounds, so I wanted my bike to roar V8 sounds as I pedal along. So I decided to make a device that measures the speed of my bike and plays car sounds, or simulates the sound of engine revs to the speed of my bicycle. It outputs the sounds through a speaker mounted in the back.



I started with the prototype: I built a 6 speed manual gearbox with lego to vary the speed of a turning magnet sticking out of the axle and started writing the code around it to simulate the car sounds through the speakers (on the left)

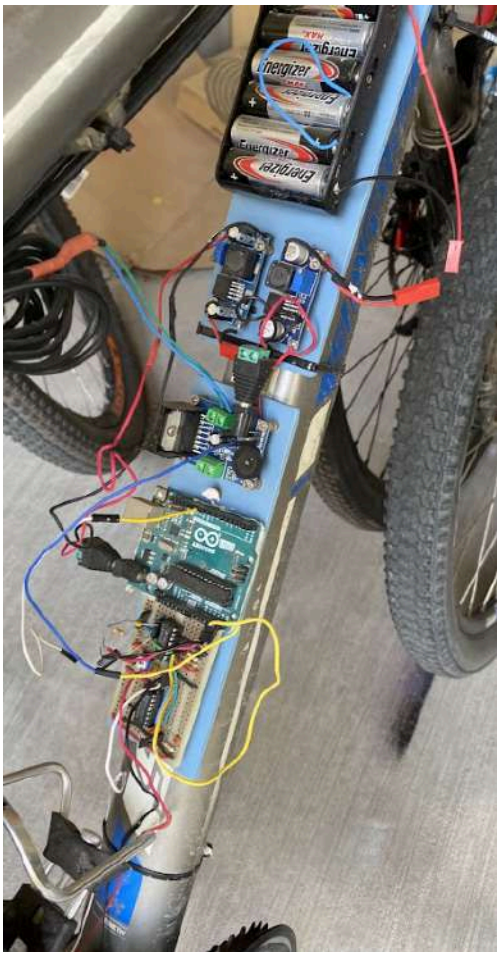
The concept is a magnet sensor that calculates the speed at which the magnet is spinning and sends it to a frequency to voltage converter chip. Then the chip sends a voltage according to the speed to the arduino uno. The arduino uno reads the voltage input and it changes the car's sound frequency and sends it to the speaker to blast. The decision to use voltage to frequency converter, was to learn about this IC I stumbled onto, at the same time it removed some coding complexity from the arduino. This also helped as the speed of the arduino was barely enough to play the sounds.

Then It was time to take it to the bike.
I 3D printed a Magnet sensor holder and stuck a couple of magnets at an equal distance from each other around the bike wheel in order to measure the speed of the wheel rotation.



The signal from the magnet sensor comes to a 4050BP IC just to clean the signal, and is connected to the LM290 frequency to voltage converter.

Some soldering later, and I had the electronics ready to be installed on my bike.
I did not do them on a milled pcb, as there was not too much complexity, and it was just faster.

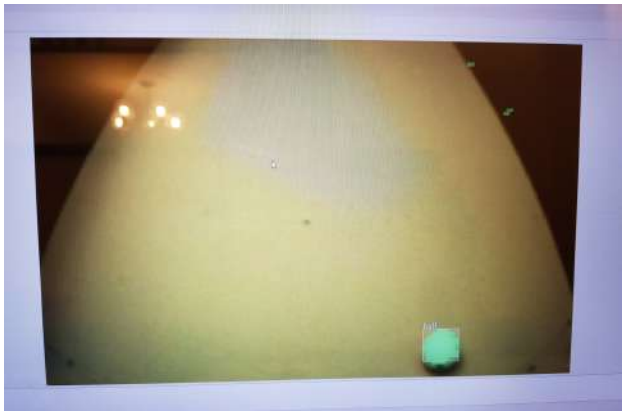


After 3D printing some support for the components as well as figuring out the power side of the project, I had my bike sound emulator ready.

Penalty Shot Blocker Project:

In a certain sense this was the project that propelled me in this hobby of coding microcontrollers and 3d printing. Winning 1st place in the Qatar National Scratch Coding Competition (QNSCC) Robotics edition was just too overwhelming at this age. I was in 8th grade.

The competition theme was Fifa related, so it was just a no brainer to invent a penalty shot blocker. The general idea was to use a ping pong ball as the football. A Pixy camera was mounted over the play area, and it would output the x,y coordinates of the ball.

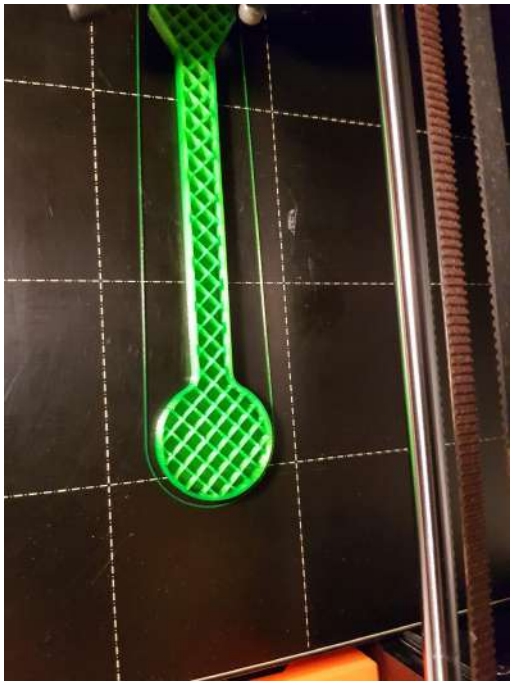


These coordinates were sent to the arduino, the tricky part as I remember was to adjust the coordinates as the camera sees the ball going from top to bottom. I had the help of my technology teacher and my Dad (my Mentors) to help me in these calculations, to be able to continuously interpolate the path and move the blocker to the intersection of the trajectory to the goal line.

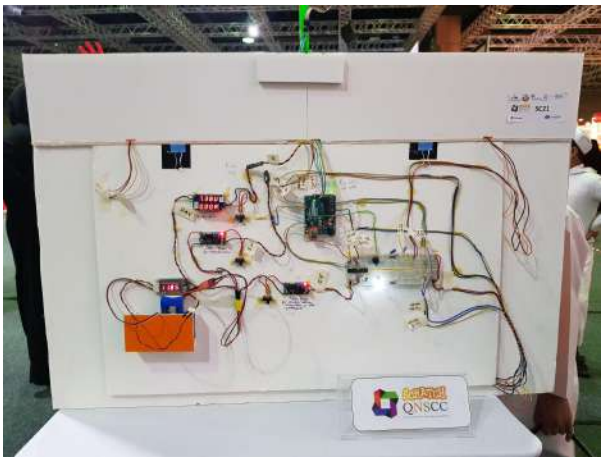
Additionally as I was using a 2 joint arm I had to find a way to know the 2 angles of the servo to move the arm to the correct position. I segmented the goal line to some 20 parts and by trial and error determined the 2 servo angles required for each of the 20 points, that was around 1.5 cm to cover each position.

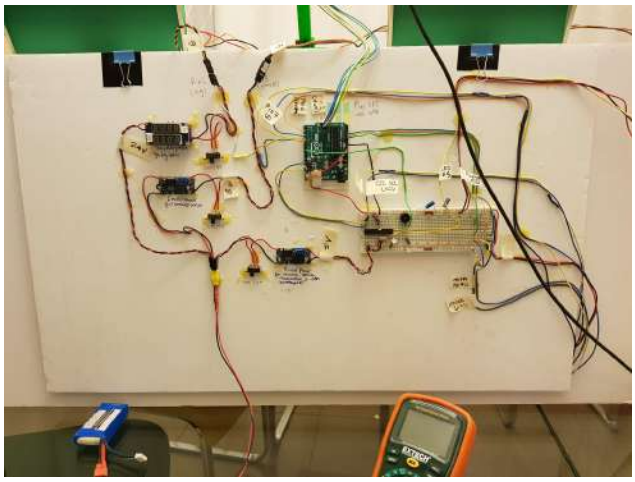
Later while discussing with my coach (teacher / mentor) there was some high level mathematics involved to be able to get the 2 servo angles calculated from the ball intersection to the goal line; which I later got to know as Inverse Kinematics! (IK), which I used in the final version.

Kindly note the ball is assumed to be on the ground always for ease of project, else it would have needed some more cameras and 3d interpolation which puts an 8th grader's brain out of order.
Nowadays, I would love to redo the project with the 3D taken into account.
Anyway back to the project; all the parts were 3D printed on my prusa MK3, they were designed with the help of my Dad on Fusion 360 and sliced using Prusa Slic3r.

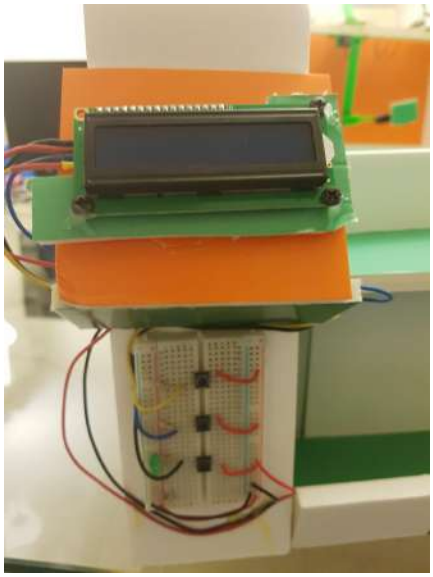


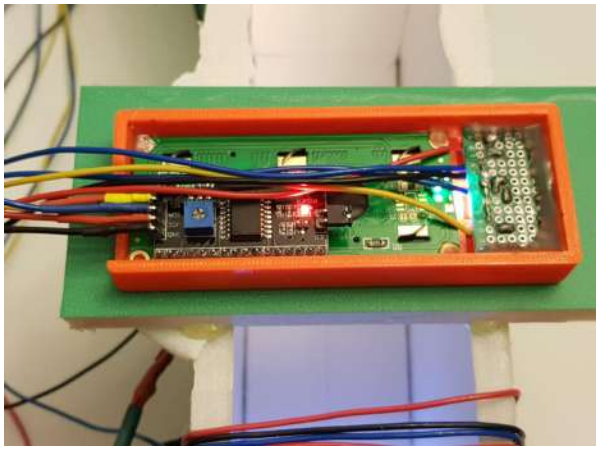
We kept all the wiring exposed on the back of the playing area to ease the explanation to the jury.

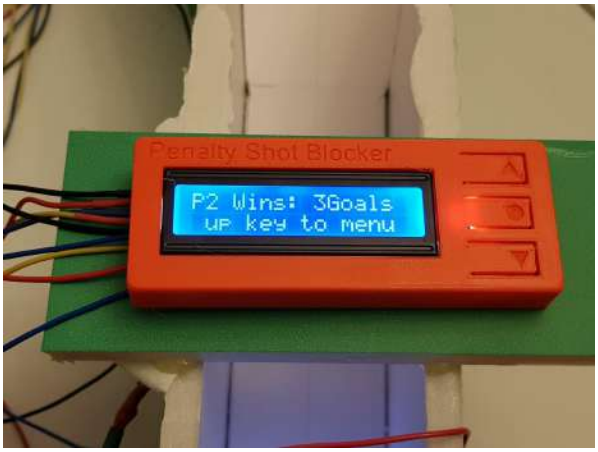




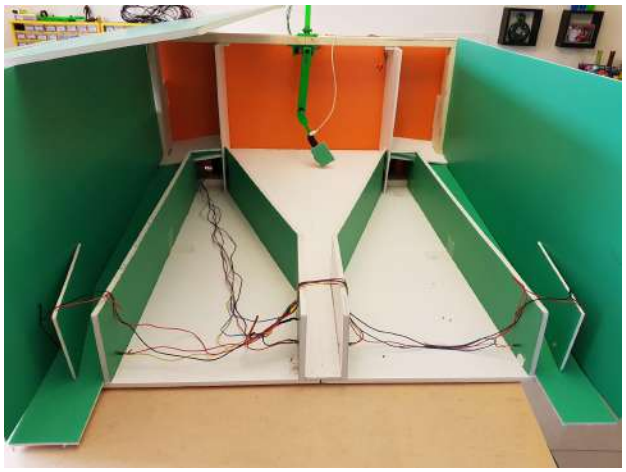
On the front there was a small menu with 3 buttons, the buttons were integrated in the case and 3D printed also. The screen was connected by i2c to the Arduino.
We could not go too fancy as all the coding was required by the rules to be done in Scratch code.

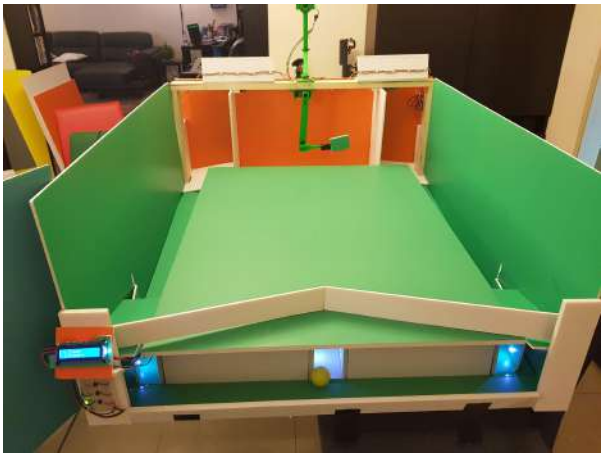




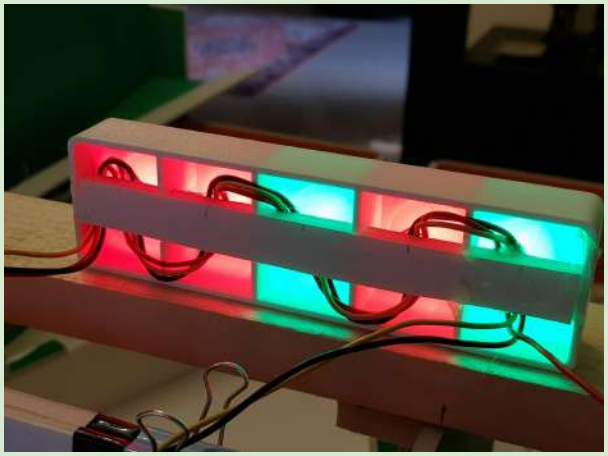
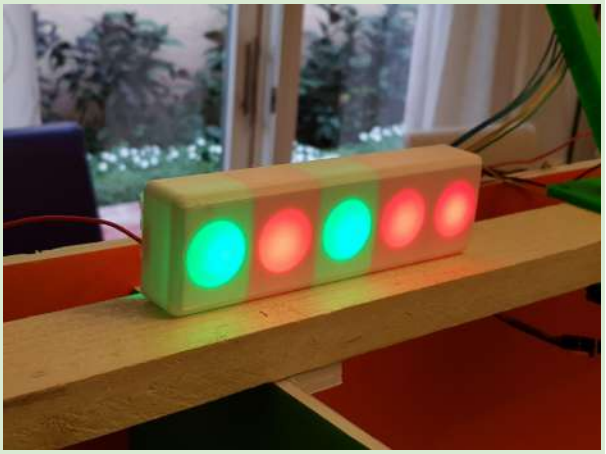


To detect a goal or out (miss), the ball bath if in the goal would follow a ramp that narrows its width and had a light sensor mounted on the wall of the ramp with a led shining on the light sensor from the other wall. When an object (the ball) intersects the led light shining on the light sensor, It would be counted as a goal. Same applies for the balls going out on the left and right side of the goal, with 2 ramps and 2 sets of light sensors and leds for detecting the no goal balls.





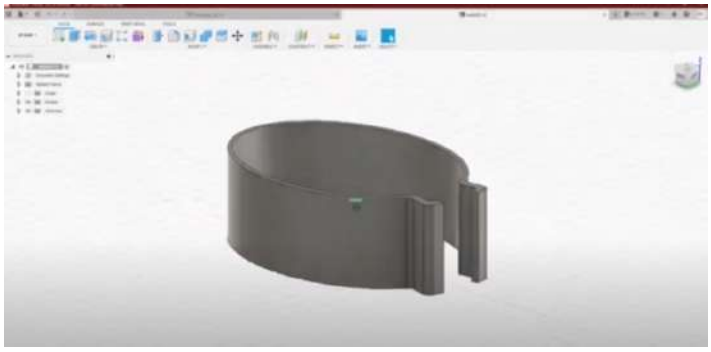
The addressable LED lights on top change color to green or red to mark a goal or out as the players compete.

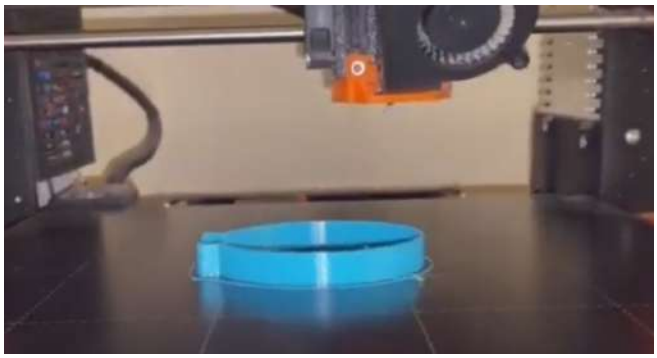



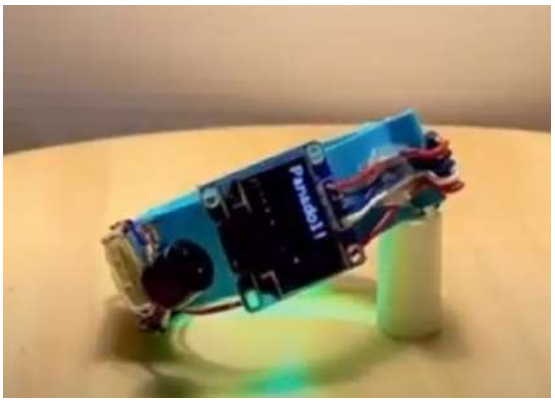
Medical Smartwatch Project:

This project was for the Middle East “Cre@school” creativity and innovation festival competition. I decided to create a medical smartwatch capable of real-time monitoring heart rate, which also tracks and sends reminders to the patient to take the correct medicine at the correct time. Additionally it tracks the ordered and consumed medications, and automatically orders the needed medications online. (this software was a mockup) The device aimed to bridge the gap between patients and doctors, ensuring immediate response in emergencies and aiding in medication adherence. The idea for this innovation came to me while observing my grandparents struggle with the timing of taking their pills.

The watch relied on a Arduino Nano wired to an LCD screen, a small motor for vibrations, a buzzer, and a heart rate sensor.
The base of the watch was modeled in Fusion 360 and 3D printed. Once put together, the prototype won me the first place award in the competition.







Autonomous Firefighter Project:

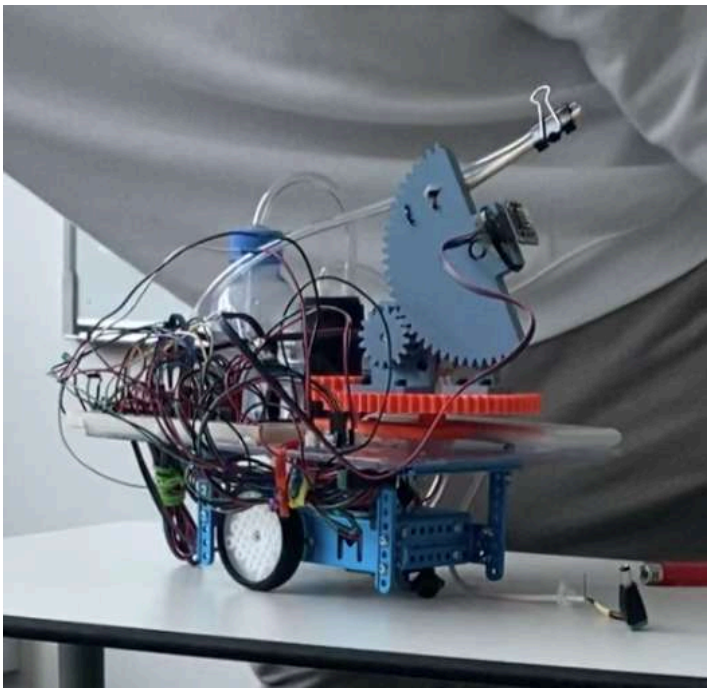
During my Spanish class, which happens to be an optional fourth language course at our school, the assignment for the next session (two days later), was to present a tech innovation orally in Spanish.

The teacher challenged me to create and build my own innovation
I decided to make an autonomous mobile, heat seeking, fire fighting robot with a moving turret pumping water putting off small flames.

Having just 2 days to come up with a working robotic project, I went into over-drive mode.
Scouring what i had lying around from unfinished projects or projects on my long to do list, I got hold of the following items:

I had a thermal camera sensor (AMG8833 IR Thermal Camera) which I bought for no project in mind, just wanted to test and see how it worked as it was quite fascinating and interesting reading about it, so I just had to but it.

2x NEMA17 stepper motors and their corresponding stepper drivers. (any tech kid would have stepper motors lying around)



Dismantled my brother's Mbot and salvaged only the base with 2 motors for moving and steering.
Found an H-bridge to control the motors of the base.
I was able to find in the local shop a small diaphragm pump (usually for aquariums) and hooked it up to another motor driver.

Turned on Fusion 360 to model the two gears with the NEMA 17 steppers incorporated to act as a turret, one stepper would turn the turntable base around the Z, and on the turntable another vertically mounted half gear to rotate in the vertical plane.

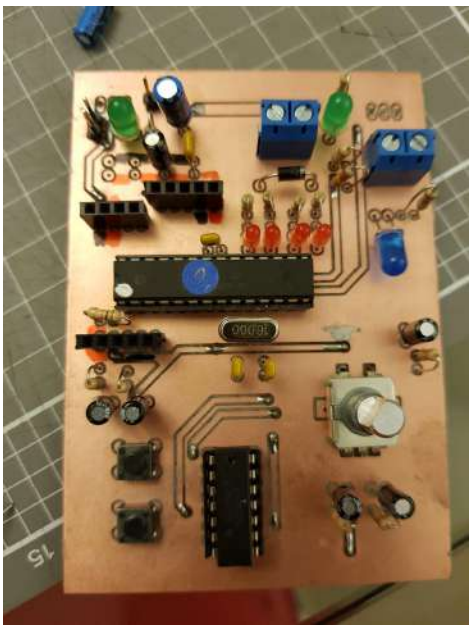
Connecting all the wires to the arduino and starting coding, while the code was rudimentary and straight forward as I was tight on time, it worked flawlessly. The robot was fully operational, although the final product was too messy, and not polished but shows what can be done in 48 hours…

Magnetic Levitator Project:

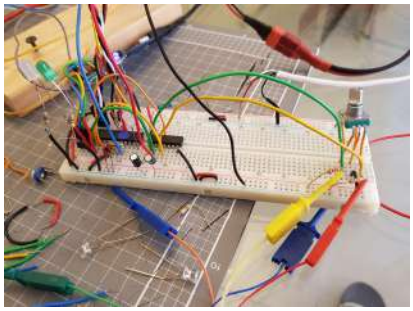
Levitation was always magical and when you see it in any form or on stage it was always mesmerizing. This along with pushing my skills and learning about new concepts was a challenge I wanted to pursue.
First was to find a magnetic Hall effect sensor with an analog output, to detect the distance of the magnet from the sensor, after researching I settled on a SS49E linear Sensor.
Next was to get an insulated copper wire from a broken treadmill motor I salvaged from my mom's treadmill when she threw it away.
Knowing the Arduino didn't have enough juice to power the electromagnet, I had to use a

transistor to channel the 12v from my RC lipo battery to the electromagnet. I burned some (>5 transistors) till I realized the power the electromagnet was pulling. Then while researching to find a solution, I realized a FET would be better and faster to switch the electro-magnet on and off, and can handle more power!

The idea was simple, the Sensor mounted on the bottom of the electromagnet would read the strength of the magnet being levitated as it gets closer or further away, and the arduino uno would do the calculations and turn the electromagnet on or off in a pulsing method. The duration of the on pulse was how much strength was needed to pull the levitating magnet.

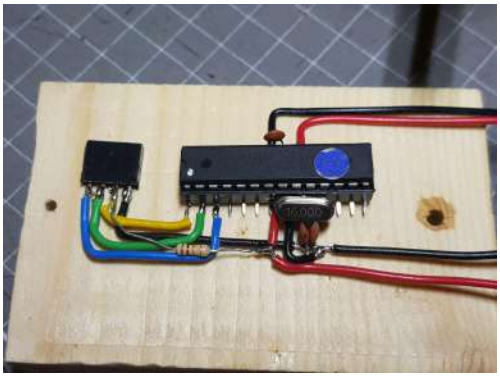


Linearly it was not possible to levitate as I was always linearly linking the pulsing or strength of the electromagnet to the magnetic sensor output, this is when I stumbled on and learned about PID in the field of motors, and it was a similar situation I had. The Proportional, Integral, Differential method would fine tune and anticipate the power needed, after many hours of tuning,



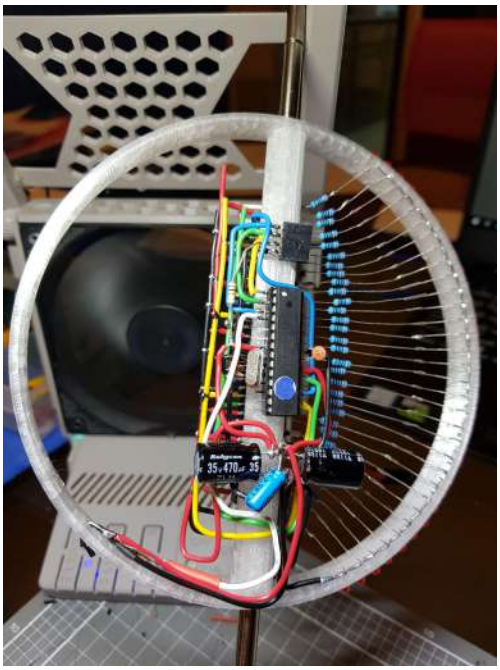
Spinning globe:

Fascination and learning new stuff was my motivation for this self project. I saw it on a cooling fan in a store with a bunch of lines and flowers, and wanted to take it a notch further at the same time have it in a spherical shape. For this project rather than encase it I did it in a fashionable way to have all the electronics visible in the spinning globe. It was actually a partial necessity as all the electronics had to be inside the spinning globe. Additionally it helped it being aerodynamic with the resistors fanned out in the method I soldered them.



An arduino uno had not many pins to drive the 32 leds I wanted to be the number of horizontal lines, the vertical lines was dictated by the switching frequency or how much I divided one turn of the globe. And an Arduino Mega would be too big, a teensy would fit, but

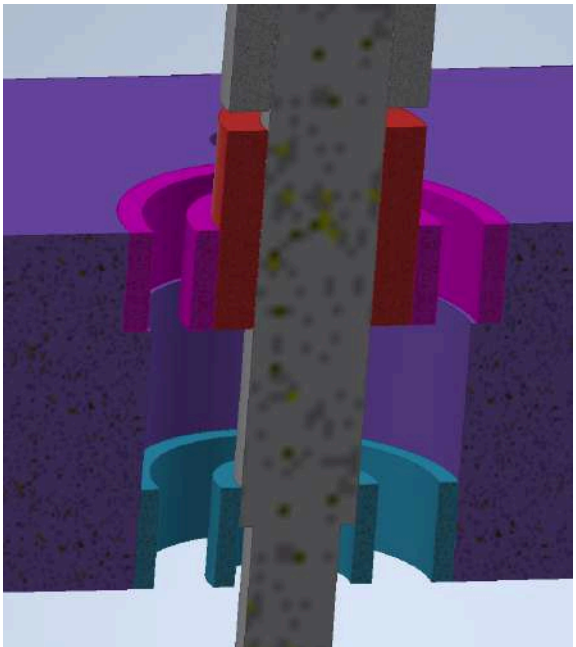
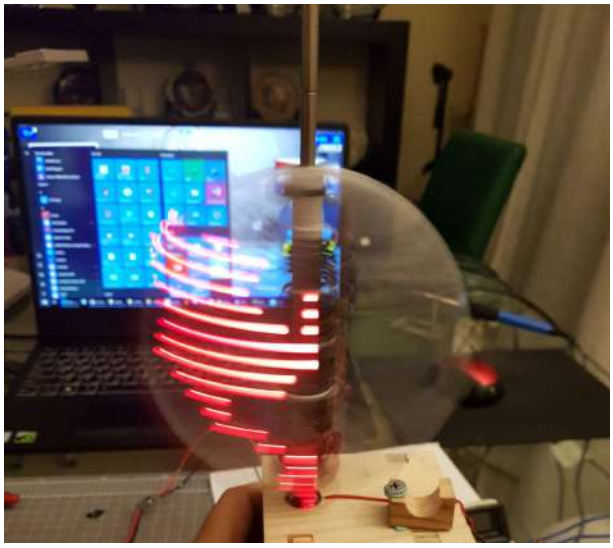
needed more challenge and learning, this is also where I stumbled on shift registers and the concept of multiplexing.



So I decided to use 4 x SN74HC595 8-Bit Shift Registers connected to 32 leds, and everything had to be bare bones, no PCB.

A simple magnet fixed on the outer C-structure would trigger the magnetic sensor on the globe to signal a full turn, the duration for this turn would be used to calculate the duration of each vertical line on and off to simulate the image.

Channeling the power to the globe was tricky, I didn't want to have to mount a battery and recharge it, but to keep it powered from a 12V wall adapter. I used 2 bearings on the lower part to support the shaft, which also helped with straightening the globe while rotating. The top bearing was oversized to the shaft diameter, and a plastic 3d printed spacer was used to insulate the shaft from the top bearing. So I had the -ve on the lower bearing and =ve on the upper bearing, and it was easy to tap off from inside the spinning globe.

It was twitchy, resetting all the time; so a couple of big capacitors helped to stabilize the power when there was non contact through the bearing while spinning.
In real life the image is crystal clear and beautiful.

Oh and I 3D printed the lower gears, and used flex filament to 3d print a belt; inventive ways to use the 3D printer to the max!